DEPARTMENT OF COPUTER SCIENCE & ENGINEERING

SUBJECT: INFORMATION THEORY CODING

ASSIGNMENT 2

Q1: Describe viterbi algorithm & maximum likelihood decoding in detail.

Q2: The parity bits of (7,4) Linear systematic block code are generated by

$C_5=d_1+d_3+d_4$

$C_6=d_1+d_2+d_3$

$C_7=d_2+d_3+d_4$

Find the generator matrix G & parity check matrix H for this code. Draw the encoder circuit.

Q3: Discuss the procedure for generating systematic cyclic code. Draw and explain the systematic cyclic encoder for a (15,9) cyclic code with generator polynomial: